[ENTER COMPANY NAME]

QUALITY MANAGEMENT SYSTEM

STANDARD OPERATING PROCEDURE

Sterilization Process Management and Validation

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Applicable Regulatory Standards	ISO 13485:2016 Clauses 6.4, 7.5.5, & 7.5.7 ISO 11135 / ISO 11137 / ISO 17665 FDA 21 CFR Part 820.75 (Process Validation)

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1.0 Purpose

This Standard Operating Procedure (SOP) defines the operational mandates and validation requirements to guarantee that medical devices required to be labeled as "STERILE" achieve an absolute, documented Sterility Assurance Level (SAL) of no less than 10^{-6}. The procedure enforces strict parametric verification, bioburden surveillance controls, and ongoing dose audits to maintain a validated microbial eradication state.

2.0 Scope

This procedure applies to all products commercialized in a sterile configuration, including contract-sterilized lots and intra-facility operations. It covers cleanroom bioburden management, sterilization cycle monitoring, biological/chemical indicator placement, contract sterilizer oversight, and product adoption models.

3.0 Responsibility and Authority

3.1 Operations / Sterilization Engineer

Responsible for designing validation protocols, assessing load configurations, performing routine parametric data checks for production batches, and monitoring the technical status of contract facilities.

3.2 Microbiology / Quality Control Laboratory Team

Responsible for performing routine pre-sterilization bioburden testing, assessing environmental particle monitoring trends, incubating exposed biological indicators (BIs), and conducting formal sterility testing.

3.3 Regulatory Affairs and Quality Assurance Manager

Holds final authority to sign off on product sterilizer adoptions, approve validation reports, and execute release sign-offs for final sterile distributed inventories.

4.0 Procedure

4.1 Core Sterilization Modalities Requirements

The organization utilizes specific, standard external validation matrices depending on product raw materials and embedded structures. Every cycle parameters must operate within rigid, fully validated limits:

Sterilization Modality	Governing Technical Standard	Critical Controlled Cycle Parameters
Ethylene Oxide (EO) Gas	ISO 11135 Lifecycle Matrix	Gas concentration (mg/L), pre-conditioning relative humidity ($RH\%$), chamber temperature, gas dwell exposure duration, exhaust flush flammability boundaries.
Gamma Radiation / E-Beam	ISO 11137 Series (VDmax / Method 1)	Minimum target absorbing dose (kGy), maximum dose threshold limits, conveyor velocity tracks, dosimeter positioning configurations.
Moist Heat / Steam	ISO 17665 Architecture	Chamber steam pressure (bar), equilibrium exposure temperature ($121^\circ ext{C}$ or $134^\circ ext{C}$), thermal saturation timing lines, air removal vacuum depth.

4.2 Pre-Sterilization Bioburden Monitoring

Quality Control must track bioburden baseline values before exposing devices to sterilization. Random samples must be pulled from production lots at a defined frequency (e.g., monthly or every 10 production lots) before packaging. Microbial counts must remain below established alert and action limits (e.g., Alert: 50 CFU, Action: 100 CFU). If an action limit is breached, the lot must be quarantined and investigated for environmental controls or raw material contamination.

> **CRITICAL PROCESS WARNING:** For radiation-sterilized items, a significant spike in pre-sterilization bioburden may compromise the validated $25 ext{ kGy}$ or $15 ext{ kGy}$ baseline dose profile, rendering the SAL invalid and triggering an immediate CAPA investigation.

4.3 Validation Protocol (The IQ / OQ / PQ Matrix)

New configurations or equipment setups must pass a formal structural validation process before being used for commercial products:

Installation Qualification (IQ):
Verify that physical utility frameworks, gas injection lines, vacuum pumps, and temperature controller modules are installed and calibrated according to engineering specifications.

Operational Qualification (OQ):
Demonstrate that the chamber operates reliably and maintains uniformity across full operating ranges (e.g., vacuum depth, pressure holds, and temperature distribution) without product loading.

Performance Qualification (PQ):
Conduct consecutive cycles (typically three consecutive successful runs) using real product in worst-case loading configurations. This stage must prove temperature uniformity and establish a Sterility Assurance Level (SAL) of 10^{-6} using biological challenges (such as *Bacillus atrophaeus* or *Geobacillus stearothermophilus* strips).

4.4 Parametric Release and Routine Monitoring Operations

Each production load must be monitored using calibrated physical sensors, chemical indicators (CIs), and biological indicators (BIs) distributed throughout the load. A batch can clear sterilization release only if all physical sensor charts confirm that critical variables stayed within validated limits, and chemical indicators show appropriate color transitions. BIs must be incubated for the required time with no signs of microbial growth.

4.5 Product Modification and Re-Validation Triggers

Changes to product components, raw materials, physical geometry, packaging materials, density, or load configurations require evaluation through the change control pipeline. The Sterilization Engineer must determine if full validation testing is required or if the modified device can be formally adopted into an existing validated sterilization load profile.

5.0 Documentation Archiving Framework

All sterilization processing charts, dosimeter mapping data, biological indicator incubation logs, and validation technical files must be retained within the Device History Record (DHR). These records must be archived for the commercial lifetime of the product family plus an additional five (5) years, and must remain legible and available for inspectorate review.

6.0 Associated Reference Materials

- **SOP-QA-002:** Change Control System
- **SOP-MF-011:** Environmental and Cleanroom Control SOP
- **Form-MF-010A:** Sterile Batch Parametric Release Checklist
- **Form-MF-010B:** Bioburden Trending Monitoring Log

7.0 Revision History Log

Revision Level	Effective Date	Author / Originator	Detailed Description of Modifications
00	[Date]	[Name], Process Validation	Initial baseline release of formalized Sterilization Process Management and Validation Standard Operating Procedure.